

Jinqiang Yu

Sydney NSW 2000, Australia

✉ jinqiang.yuuu@gmail.com | [in linkedin.com/in/jinqiang-yu-404bb0187](https://www.linkedin.com/in/jinqiang-yu-404bb0187) | 🎓 Jinqiang Yu

Summary

Research Engineer at TikTok specializing in pre-training and post-training of LLMs, with expertise in explainable AI, NLP, and multimodal LLM models. Lead author in top AI venues (AAAI, JAIR) and reviewer for AAAI/IJCAI. Stay updated on cutting-edge AI techniques like generative AI and LLMs. Proven track record of translating research into SOTA models powering business scenarios.

Education

Monash University

Melbourne, Australia

PhD in Artificial Intelligence

Feb 2021 - Jul 2024

- **Thesis Topic:** Explainable AI for Machine Learning
- **Thesis Description:** The PhD project aims to deal with explainable AI and machine learning problems, including training interpretable machine learning (ML) models, generating accurate and concise explanations to explain ML models in various domains like NLP, CV, healthcare and general classification tasks, and applying explainable AI for ML, in particular LLMs, such improving LLM performance and enabling LLMs to generate explainable outputs.

Monash University

Melbourne, Australia

Master of Information Technology

Mar 2019 - Dec 2020

- Graduated with H1
- **Thesis Topic:** Computing optimal interpretable machine learning models.
- **Thesis Description:** The thesis focuses on interpretable models, aiming at developing advanced approaches to computing machine learning models that are both accurate and interpretable.

Experience

TikTok

Sydney, Australia

Research Engineer

Jul 2024 - Present

- Lead LLM pre- and post-training research to translate empirical insights into business gains – covering data selection/collection, evaluation, training optimization, and prompt engineering.
- Based on my pre-training research findings, produced a state-of-the-art LLM that has been used in 6 high-impact business downstream tasks such as moderation.
- Explore and deliver post-training tricks, benefiting business downstream scenarios.
- Lead juniors and lead cross-functional research projects; deliver tech talks and internal best-practice guides.

OPTIMA

Melbourne, Australia

PhD Researcher

Apr 2021 - Present

- Engaged in research projects at OPTIMA, which focuses on explainable AI for machine learning.
- Projects' topics include generating interpretable ML models and computing accurate and concise explanations, aiming at developing methods to help users understand and explain ML model inferences and predictions, and applying explainable AI to improve ML performance in diverse domains, especially LLMs.
- Collaborated with members across teams to push projects forward. Delivered tech sharing with colleagues.

Professional Projects

Pre-training to produce SOTA LLM base models

Sydney, Australia

TikTok

Jul 2024 - present

- Led an LLM research project on pre-training, aiming to produce a state-of-the-art (SOTA) LLM base models for downstream tasks. Dive into four main direction research: data, training, model, evaluation.
- Based on the research findings, produced **SOTA base models** for downstream tasks, improving task metric by 10% over previous baselines. The proposed models have been used in 6 downstream scenarios, such as moderation and search.
- Built an end-to-end pretraining pipeline over trillions tokens, leveraging mixture curation, deduplication, quality filtering, and domain balancing.
- Improved validation set construction: increased downstream correlation, providing effective approaches to fast evaluation, model selection and data selection.
- Conducted ablations across model scales (e.g., 1B–32B), and tokenizer variants; identified optimal data/model/training trade-offs for downstream.
- **Expertise:** Data Selection/Collection, Pre-training, Post-training, Scaling Law, Evaluation, Inference, (Multimodal) LLM, MoE.

Post-training to boost LLM downstream performance

Sydney, Australia

TikTok

Jul 2024 - present

- Led LLM research projects on post-training to boost model performance in downstream tasks.
- Identified more effective reasoning-learning paradigms, leading to increasing downstream task performance by 10% on downstream tasks.
- Designed robust hyperparameter tuning and validation-set construction methods, ensuring optimal model performance.
- Built data curation and selection strategies, including an AI labeling agent for high-quality labels and data mixture pipeline.
- **Expertise: Post-training, Data Selection/Labeling, Prompt Engineering, Agent, Reasoning, Evaluation, Inference, (Multi-modal) LLM.**

Applying Explainable AI in Machine Learning and LLMs

Melbourne, Australia

Monash University/OPTIMA

Oct 2023 - Aug 2024

- Conducted research on LLMs with the use of explainable AI. Aiming to improve LLMs’ performance, help others understand the models’ behavior, and build trust of the models.
- Developed approaches to improving the performance of LLMs with the use of Parameter-Efficient Fine-Tuning (PEFT) and Prompt Engineering given explanations.
- Developed methods to generate more explainable outputs of LLMs for different users with the use of Instruction Fine-Tuning and RL.
- **Expertise: Machine Learning, LLMs, Generative AI, Prompting Engineering, PEFT, Instruction Fine-Tuning, RL, Explainable AI.**

Computing Succinct and Accurate Explanations for ML

Melbourne, Australia

Monash University/OPTIMA

Feb 2021 - Apr 2023

- Conducted research on generating more accurate and concise post-hoc (*abductive*) explanations answering why the prediction is made and (*contrastive*) explanations targeting why not the prediction is made.
- Developed approaches to mining background knowledge and applying them to generating accurate and concise explanations in diverse domains.
- Developed approaches to generating feature attribution indicating the contribution of a feature.
- **Expertise: Machine Learning, Data Mining, Explainable AI.**

Selected Publications

Anytime Approximate Formal Feature Attribution

Jinqiang Yu Graham Farr, Alexey Ignatiev, Peter J. Stuckey

arXiv preprint arXiv:2312.06973, 2023

On Formal Feature Attribution and Its Approximation

Jinqiang Yu, Alexey Ignatiev, Peter J. Stuckey

arXiv preprint arXiv:2307.03380, 2023

From Formal Boosted Tree Explanations to Interpretable Rule Sets

Jinqiang Yu, Alexey Ignatiev, Peter J. Stuckey

29th International Conference on Principles and Practice of Constraint Programming (CP), 2023

Eliminating the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background Knowledge in the Context of Formal Explanations

Jinqiang Yu, Alexey Ignatiev, Peter J. Stuckey, Nina Narodytska, Joao Marques-Silva

37th AAAI Conference on Artificial Intelligence (AAAI), 2023

Skills

Programming Languages

Python, Java, R, C++, MATLAB

DS/ML/AI Technologies

Pre-training & Post-training

Domains - NLP, CV, LLM, Generative AI, Explainable AI

ML/DL/GenAI Models - (Multimodal) LLM, CNN, Diffusion model, GAN, VAE, etc

RAG, RLHF, and Prompt Engineering

Cleansing and Wrangling - Pandas, Numpy, NLTK

Visualization - Matplotlib, Seaborn

Big Data Management - MongoDB, SQL, Spark

Miscellaneous Technologies

Github, Docker, AWS, Slurm, Linux

Scientific Activities & Awards

2024-2026 PC member of the AAAI Conference on Artificial Intelligence

2024 PC member of the International Joint Conferences on Artificial Intelligence (IJCAI-2024)

2020 **Best Paper Award:** My paper “Computing Optimal Decision Sets with SAT” has been selected for the Best Paper Award for the CP/ML Track of CP 2020.